$\varphi = 10 - 4t^2 + t^3$

$\omega = 12 c^{-1}$

$t - ?$

$\varepsilon - ?$

$\omega = \frac{d\varphi}{dt} = -8t + 3t^2$

$12 = -8t + 3t^2$

$3t^2 - 8t + 12 = 0$

$D = 64 - 4 \cdot 3 \cdot 12 \to 208$

$t_1 = 3,73$

$t_2 < 0$

$\varepsilon = \frac{d\varphi}{dt} = -8 + 6t$

$\varepsilon = -8 + 6 \cdot 3,73 = 14,38 \frac{рад}{с^2}$

$R = 10 см$

$\varphi = a + 2t^2 + t^3$

$t = 2c$

$\omega - ?$

$V - ?$
$a_n - ?$
$a_\tau - ?$
$\varepsilon - ?$
$a - ?$

$\omega = 4t + 3t^2$

$\omega(t=2c) = 8 + 12 = 20 \frac{рад}{с^2}$

$V = \omega R$

$V = 20 \cdot 0,1 = 2 \frac{м}{с}$

$\varepsilon = 4 + 6t$

$\varepsilon = 16 \frac{рад}{с^2}$



$a_t = \varepsilon R$

$a_\tau = 1,6 \frac{м}{с^2}$

$S = ct^3$

$c = 0,1 \frac{см}{с^3} = 10^{-3} \frac{м}{с}$

$S = 10^{-3} t^3$

$R = 2 см = 2 \cdot 10^{-2} м$

$V = 0,3 \frac{м}{с}$

$a_n - ?$

$a_t - ?$

$a_n = \frac{V^2}{R}$

$a_n = \frac{0,3^2}{2 \cdot 10^{-2}} = 4,5 \frac{м}{с^2}$

$V = \omega R$

$V = 3 \cdot 10^{-3} t^2$

$a_t = 6 \cdot 10^{-3} t$

$a_t = 6 \cdot 10^{-3} \cdot 10 = 0,06 \frac{м}{с^2}$

$\varepsilon = 4 + 6t$

$\omega (t = 5c) = 10 \frac{рад}{с}$

$\varphi_0 = 0 рад$

$\varepsilon (t) - ?$

$\omega (t) - ?$

$\omega = \int \varepsilon \, dt + \omega_0$

$\omega = 4t + \frac{6t^2}{2} + \omega_0 =$

$4t + 3t^2 + \omega_0$

$\omega (t) = 4t + 3t^2 - 53$

$\varphi (t) = \int \omega \, dt + \varphi_0$

$\varphi (t) = \frac{4t^2}{2} + \frac{3t^3}{3} - 53t$

$d = 300$ мм | $3 \cdot 10^{-1}$

$V = 35 \frac{м}{с}$

$\nu_1 = 1400 \frac{об}{мин}$ | $23,3 \frac{об}{с}$

$\nu_2 = 2800 \frac{об}{мин}$ | $46,6 \frac{об}{с}$

$V_1 - ?$

$V_2 - ?$

$\nu = \frac{1}{T}$

$V = \omega R$

$\omega = \frac{\varphi}{t} = \frac{2\pi}{t}$

$V = 2\pi \nu \cdot R = \pi d \nu$

$V_1 = 3,14 \cdot 23,3 \cdot 0,3$

ДИНАМИКА

Категории сил:

I. Электромагнитное (заряды)

II. Гравитационные (масса)

III. Ядерное (ильное)

IV. Слабое (объединение частиц, наук. в ядерной)

Способы отыскания:

I. силовой

II. Энергетический

Сила тяжести

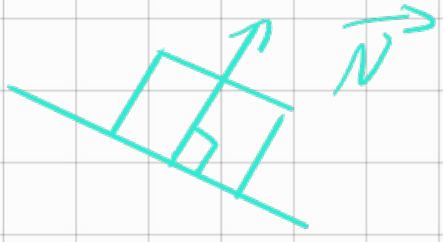
$$F = G\frac{m_1 m_2}{R^2}$$

$$G = 6{,}67 \cdot 10^{-11}$$

$$F = G\frac{M_3 m}{R_3^2} = g$$

$$g = 9{,}8$$





$m = 50\ кг$

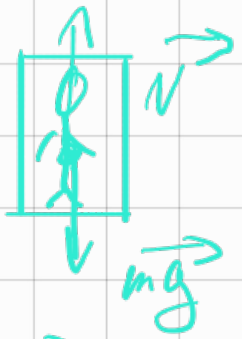
$g = 10\ \frac{м}{с^2}$

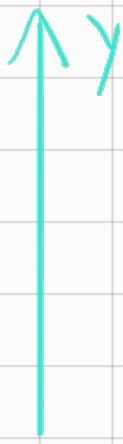
$a = 1\ \frac{м}{с^2}$

↑

$p - ?$

↓





$$\Sigma \vec{F} = m\vec{a}$$

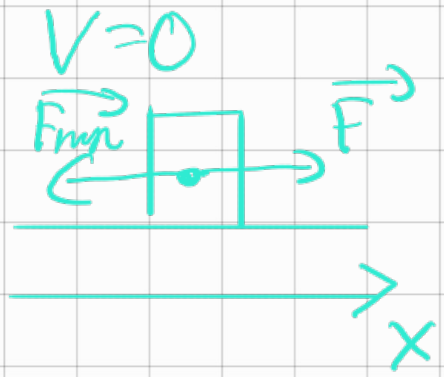
$$\vec{N} + m\vec{g} = m\vec{a}$$

$$y: N - mg = ma$$

$$N = m(g+a)$$

$$P = N = 50кг\,(10+1)\frac{м}{с^2} = 550Н$$

ТРЕНИЕ ПОКОЯ

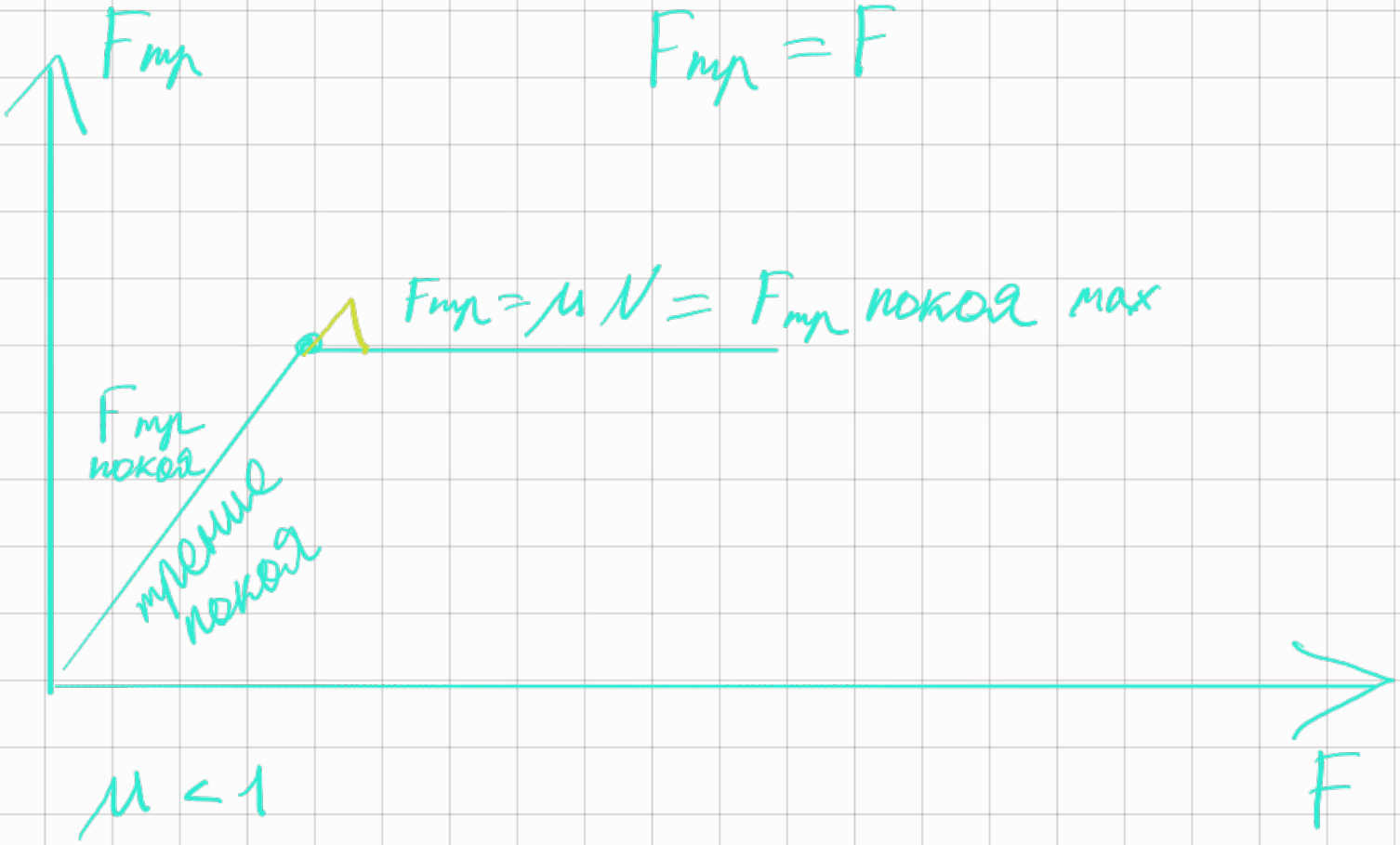


$$\sum \vec{F} = m\vec{a}$$

$$\vec{F}_{тр} + \vec{F} = m\vec{a}$$

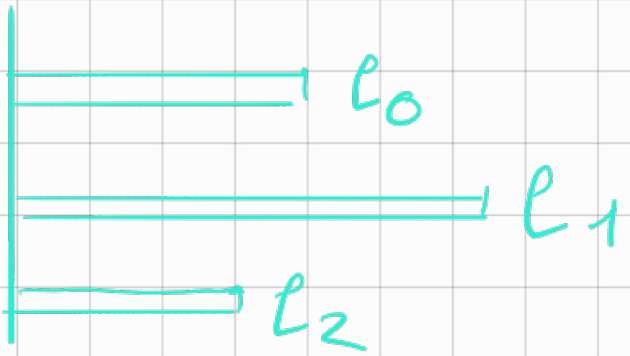
$$-F_{тр} + F = ma = 0$$

$$F_{тр} = F$$



СИЛА УПРУГОСТИ

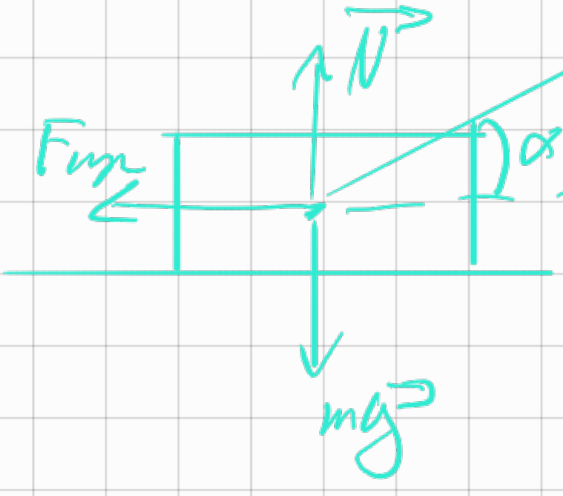
$$\vec{F} = -k\vec{x}$$



$x = l_1 - l_0$
$x = l_2 - l_0$

k – коэф. жесткости

$F_{apx} = \rho \cdot g \cdot V_{погр}$



$\sum \vec{F} = m\vec{a} = 0$

$\rightarrow x \quad \vec{F} + m\vec{g} + \vec{N} + \vec{F}_{тр} = 0$

$X: \; F\cos\alpha - F_{тр} = 0$

$y: F\sin\alpha - mg + N = 0$

$N = mg - F\sin\alpha$

$F_{тр} = \mu N = \mu(mg - F\sin\alpha)$

$X: F\cos\alpha - \mu(mg - F\sin\alpha) = 0$

$F\cos\alpha - \mu mg + \mu F\sin\alpha = 0$

$F(\cos\alpha + \mu\sin\alpha) = \mu mg$

$$F = \frac{\mu mg}{\cos\alpha + \mu\sin\alpha}$$

$F = min$, if

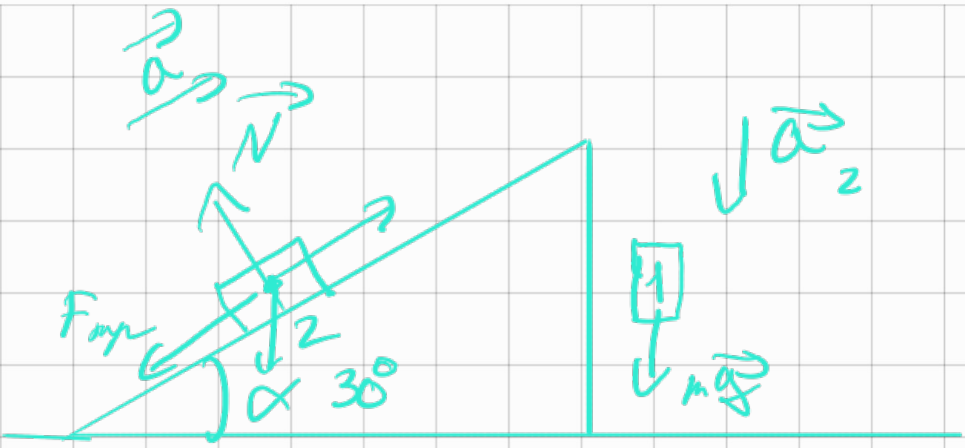
$(\cos\alpha + \mu\sin\alpha) = max$

$(\cos\alpha + \mu\sin\alpha)' = 0$

$\sin\alpha - \mu\cos\alpha = 0$

$\sin\alpha = \mu\cos\alpha \Rightarrow \mu = tg\alpha$

$\alpha = arctg\,\mu$



$\alpha = 30^\circ$





$\vec{T}_1 + \vec{T}_2 = m\vec{a}$

$T_1 - T_2 = ma$

$T_1 - T_2 \to 0 \Rightarrow T_1 = T_2 = T$

$a_1 = a_2 = a$

1 тело: $\vec{T}_1 + m\vec{g} = m\vec{a}_1$

x: $mg - T = ma$

2 тело: $\vec{T}_2 + \vec{N} + m\vec{g} + \vec{F}_{тр} = m\vec{a}_2$

$X: \quad N - mg\cos\alpha = 0$

$Y: \quad T - mg\sin\alpha - F_{тр} = ma$

$N = mg\cos\alpha \qquad F_{тр} = \mu N$

$mg - ma - mg\sin\alpha - \mu mg\cos\alpha = ma$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$F\Delta t = \Delta\vec{p}$$

Второй закон Ньютона